

Reasoning with Uncertainty

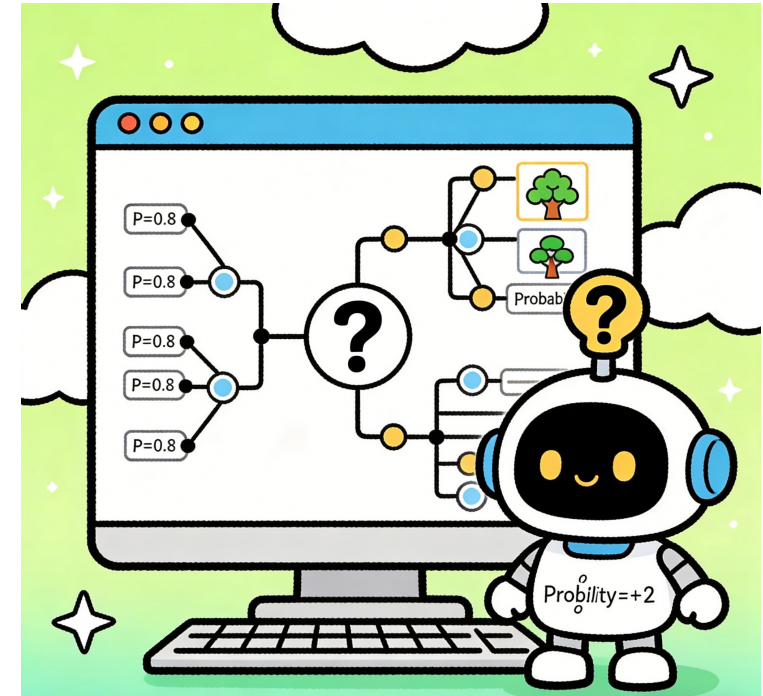
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Reasoning with Uncertainty

- Quantifying uncertainty
- Probabilistic Reasoning: Bayesian networks
- Reasoning over time



Reason with Uncertainty

Agents' plans are contingent on events in the world:

- What will be the weather in Auckland tomorrow?
- Would a COVID lockdown take place next week?
- Would the All Blacks win their next Test match?
- Would I get head or tail if I toss a coin?

Uncertainty is an important aspect of an agent's decision making:

- Agents only have partial information about the environment.
- Agents' knowledge of the truth of a statement is uncertain.
- The state of the world is inherently uncertain.

The agent's knowledge can at best provide only a **degree of belief**.

Probability

- Probability theory is a calculus of belief
E.g. To say that “There is a 80% probability of raining” expresses a belief based on past experience.
- Probability theory is the study of how knowledge affects belief.
- The probability of a hypothesis α is a scale of the agent’s belief in α in the range $[0,1]$.

Next, we will link propositional logic framework with probability to capture the above intuitions.



- Let $X_1, \dots, X_d$ be atomic propositions.

E.g. Sunny means outlook will be sunny, **Hot** means high temperature.

- The **sample space** Ω over these atomic propositions contains all possible interpretations. Each interpretation is also called a **sample**.

E.g. Samples over Sunny, Hot:

interpretations	Sunny	Hot
e_1	true	true
e_2	true	false
e_3	false	true
e_4	false	false

- A proposition describes a constraint on atoms.

E.g. 1. $\alpha_1: \neg Hot$

$$e_2 \models \alpha_1, e_4 \models \alpha_1$$

2. $\alpha_2: \neg Sunny \vee \neg Hot$

$$e_2 \models \alpha_2, e_3 \models \alpha_2, e_4 \models \alpha_2$$

Definition [Belief measure]

- For a sample $s \in \Omega$ whose features are unknown, a **belief measure over Ω** is a function $\mu: 2^\Omega \rightarrow [0,1]$ such that for any $S \subseteq \Omega$, $\mu(S)$ expresses the amount of belief in the fact that s is an element of S .
- Any belief measure must satisfy the following properties:
 - (Unit Measure) s must be a sample in Ω , i.e.,
$$\mu(\Omega) = 1$$
 - (Additivity) Suppose S_1 and S_2 are disjoint subsets of Ω . Then our belief that $s \in S_1 \cup S_2$ is the sum of our belief that $s \in S_1$ and our belief that $s \in S_2$, i.e.,

$$\mu(S_1 \cup S_2) = \mu(S_1) + \mu(S_2)$$

Definition [Probability]

Given a belief measure μ over Ω , the **probability** of a proposition α is $P(\alpha) = \mu(\{\omega \in \Omega: \omega \models \alpha\})$, i.e., it is the strength of belief that α will hold.

Example. Consider the atoms **Sunny** and **Hot**.

- Belief measure 1:

Sunny	<i>true</i>	<i>true</i>	<i>false</i>	<i>false</i>
Hot	<i>true</i>	<i>false</i>	<i>true</i>	<i>false</i>
Belief μ	0	1	0	0

This corresponds to a single interpretation in propositional logic without uncertainty.

- Belief measure 2:

Sunny	<i>true</i>	<i>true</i>	<i>false</i>	<i>false</i>
Hot	<i>true</i>	<i>false</i>	<i>true</i>	<i>false</i>
Belief μ	0.35	0.19	0.12	0.34

$$\begin{aligned}
 P(\text{Sunny}) &= \mu(\{\text{Sunny} \wedge \text{Hot}, \text{Sunny} \wedge \neg \text{Hot}\}) \\
 &= 0.35 + 0.19 = 0.54
 \end{aligned}$$

Conditional Probability

- The measure of belief in hypothesis h based on proposition e is called the **conditional probability** of h given e , written $P(h | e)$.
- The proposition e represents certain given experience.
- The probability $P(h)$ is the **prior probability** of h and is the same as $P(h | \text{true})$. It is the probability of h without any given experience.
- The conditional probability $P(h | e) = \frac{P(h \wedge e)}{P(e)}$ is the agent's **posterior probability** of h .

E.g. Prior probability $P(\text{Sunny}) = 0.54$

Evidence e : $\neg \text{Hot}$

Other knowledge: $P(\text{Sunny} \wedge \neg \text{Hot}) = 0.19$

$$P(\neg \text{Hot}) = 0.53$$

Posterior probability: $P(\text{Sunny} | \neg \text{Hot}) = \frac{0.19}{0.53} \approx 0.36$

Theorem (Properties of conditional probability)

The following hold for all propositions a and b and e :

- $P(e | e) = 1$
- If $a \wedge b$ is a contradiction, $P(a | e) + P(b | e) = P(a \vee b | e)$
- $P(\neg a | e) = 1 - P(a | e)$
- If a and b are logically equivalent, then $P(a | e) = P(b | e)$
- $P(a | e) = P(a \wedge b | e) + P(a \wedge \neg b | e)$
- $P(a \vee b | e) = P(a | e) + P(b | e) - P(a \wedge b | e)$
- Chain rule: $P(a \wedge b) = P(a)P(b | a)$
- Law of total probability: $P(a) = P(a | b)P(b) + P(a | \neg b)P(\neg b)$
- Baye's rule: $P(a | b) = \frac{P(b|a) \times P(a)}{P(b)}$

Probability Distribution

Suppose $S = \{X_1, \dots, X_m\}$ is a set of atomic propositions, we use Ω_S to denote the set of samples on S , i.e.,

$$\Omega_S = \{(l_1, \dots, l_m) \mid \text{each } l_i = X_i \text{ or } l_i = \neg X_i \text{ for } 1 \leq i \leq m\}$$

Definition [probability distribution]

- A **probability distribution** of a set of atoms S , denoted as P_S , is a function from Ω_S into $[0,1]$ such that

$$\sum_{\omega \in \Omega_S} P_S(\omega) = 1$$

We express it as $P(X_1, \dots, X_m)$.

- A **probability distribution of a set of atoms S conditioned on atoms $Y_1, \dots, Y_l$** , denoted as $P_{S|Y_1, \dots, Y_l}$, is a function from $\Omega_{\{Y_1, \dots, Y_l\} \cup S}$ to $[0,1]$ such that for any $\tau \in \Omega_{\{Y_1, \dots, Y_l\}}$

$$\sum_{\omega \in \Omega_S} P_{S|Y_1, \dots, Y_l}(\omega, \tau) = 1$$

We express it as $P(X_1, \dots, X_m | Y_1, \dots, Y_l)$.

Compact Representation of Probability Distribution

Conditional probability table (CPT)

A probability distribution can be represented as a **probability table**:

Sunny	$P(\text{Sunny})$	and	Hot	$P(\text{Hot})$
0	0.46		0	0.53
1	0.54		1	0.47

We can represent conditional probability distribution as a **conditional probability table (CPT)**.

Sunny	Hot	$P(\text{Hot} \text{Sunny})$
0	0	$0.34/0.46 \approx 0.74$
0	1	$0.12/0.46 \approx 0.26$
1	0	$0.19/0.54 \approx 0.35$
1	1	$0.35/0.54 \approx 0.65$

Summary of notations:

- Atomic propositions: $X_1, X_2, X_3, \dots, Y_1, Y_2, Y_3, \dots, Z_1, \dots$
- Sample space: Ω_S where $S = \{X_1, \dots, X_m\}$.
- Probability of proposition h : $P(h)$
- Probability of proposition h conditioned on e : $P(h \mid e)$
- Probability distribution of S : $P_S = P(X_1, \dots, X_m)$
- Probability distribution of S conditioned on $Y_1, \dots, Y_l$:

$$P(X_1, \dots, X_m \mid Y_1, \dots, Y_l)$$

- CPT for $P(X_1, \dots, X_m \mid Y_1, \dots, Y_l)$:

Y_1	...	Y_l	X_1	...	X_m	$P(X_1, \dots, X_m \mid Y_1, \dots, Y_l)$
...	...	...	...	...	...	...

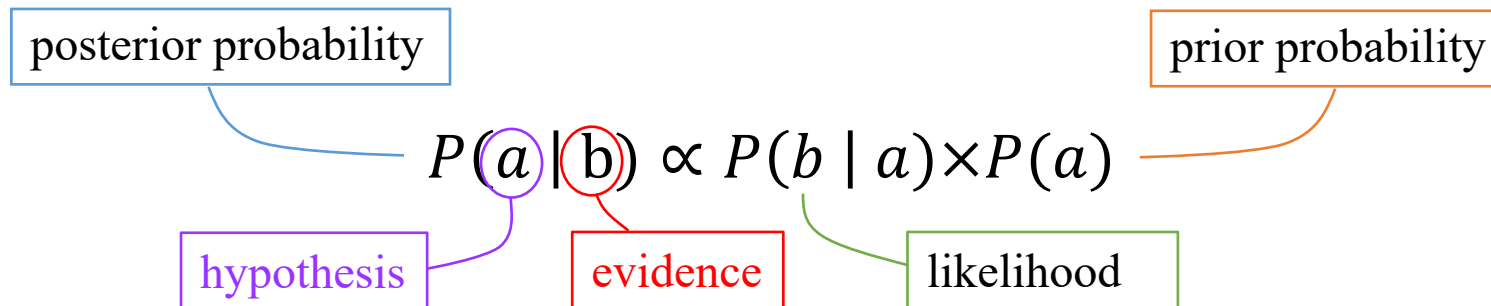
Note: **Green columns** denote variables for which (joint) probability is measured.

Inference with Probability

Recall: Baye's rule

$$P(a|b) = \frac{P(b|a) \times P(a)}{P(b)}$$

This means that $P(a|b)$ is **proportional** to $P(b|a) \times P(a)$, written as



The diagram shows the equation $P(a|b) \propto P(b|a) \times P(a)$ with several labels and arrows:

- A blue box labeled "posterior probability" points to $P(a|b)$.
- An orange box labeled "prior probability" points to $P(a)$.
- A purple box labeled "hypothesis" points to a in $P(a|b)$.
- A red box labeled "evidence" points to b in $P(a|b)$.
- A green box labeled "likelihood" points to $P(b|a)$.

or equivalently

$$\left(\frac{P(a|b)}{P(\neg a|b)} \right) = \alpha \left(\frac{P(b|a) \times P(a)}{P(b|\neg a) \times P(\neg a)} \right) \text{ for constant } \alpha$$

“Posterior_probability $\propto$ Likelihood $\times$ Prior_Probability”

Example. [infection] Suppose a test for a certain viral infection is 95% reliable for infected patients and 99% reliable for the healthy one. Suppose also that we have prior belief that 4% of patients have the virus. How to decide if the patient is infected?

Atoms:

- *Test*: the test outcome is positive.
- *Inf* : the patient is infected

Prior knowledge:

- Probability of infection $P(Inf) = 0.04$
- The test is 95% reliable for infected patients: $P(Test | Inf) = 0.95$
- The test is 99% reliable for healthy patients: $P(\neg Test | \neg Inf) = 0.99$

Task: Compute $P(Inf | Test)$.



- Write down the CPT:

Inf	$P(Inf)$	and	Inf	Test	$P(Test Inf)$
0	0.96		0	0	0.99
1	0.04		0	1	0.01
			1	0	0.05
			1	1	0.95

- By Bayes' rule:

$$P(Inf | Test) \propto P(Test | Inf)P(Inf)$$

- Thus

$$\left(\frac{P(Inf | Test)}{P(\neg Inf | Test)} \right) = \alpha \left(\frac{0.95 \times 0.04}{0.01 \times 0.96} \right) = \alpha \left(\frac{0.038}{0.0096} \right)$$

- This means that $P(Inf | Test) = \frac{0.038}{0.0476} \approx 79.83\%$

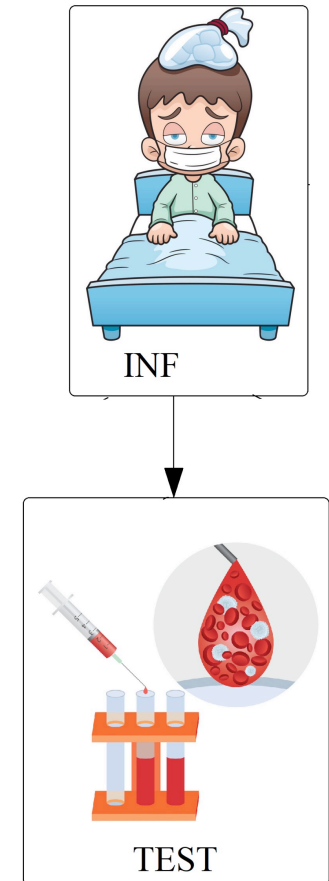
Example. [infection] The simple diagnostic problem above involves only two atoms: *Inf* and *Test*. *Inf* can be seen as a causal factor for *Test*:

- Infected patients are most likely to be tested positive.
- Uninfected patients are less likely to be tested positive.

We may represent such a causal relation with a clause: $Test \leftarrow Inf$

Note. The clause $Inf \leftarrow Test$ would not be accurate as

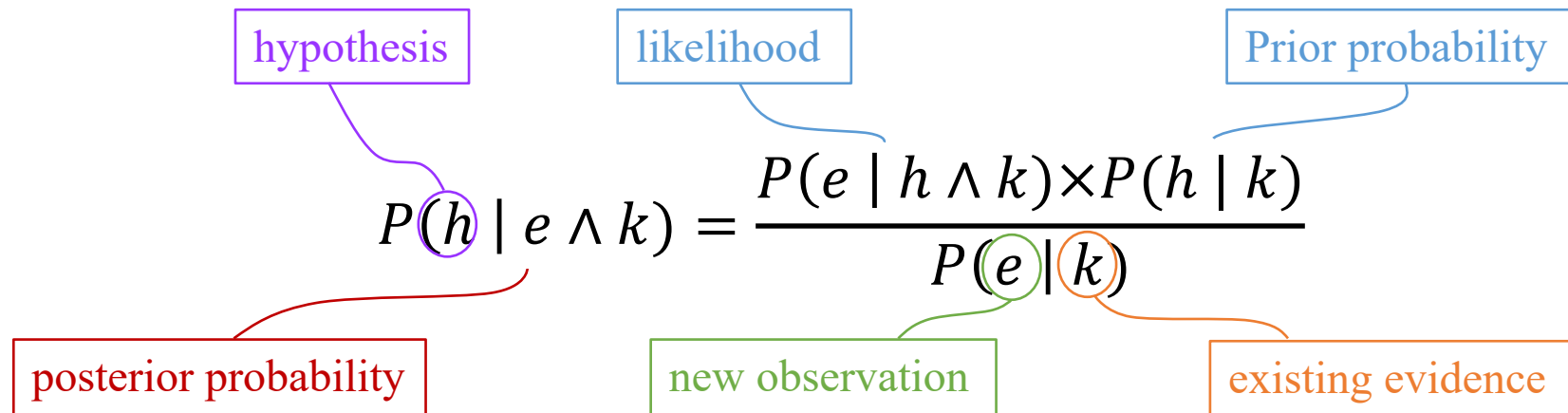
- testing positive does not guarantee infection; and more importantly
- even if the patient is infected, the fact that a patient is tested positive is not the reason for the infection of the patient.



Bayes' rule tells us how to update the agent's belief in hypothesis h as new evidence e arrives, given existing knowledge k .

Bayes' Rule with Existing Evidence

As long as $P(e | k) \neq 0$



The diagram illustrates Bayes' Rule with Existing Evidence. The central equation is $P(h | e \wedge k) = \frac{P(e | h \wedge k) \times P(h | k)}{P(e | k)}$. The components are labeled as follows: 'hypothesis' (purple box) points to h in the numerator; 'likelihood' (blue box) points to $P(e | h \wedge k)$; 'Prior probability' (blue box) points to $P(h | k)$; 'posterior probability' (red box) points to $P(h | e \wedge k)$; 'new observation' (green box) points to e in the denominator; and 'existing evidence' (orange box) points to k in the denominator.

$$P(h | e \wedge k) = \frac{P(e | h \wedge k) \times P(h | k)}{P(e | k)}$$

This means that

$$P(h | e \wedge k) \propto P(e | h \wedge k) \times P(h | k)$$

Example. [infection] Suppose a patient has been tested positive for the viral infection. Experience tells that 70% of infected patients and 30% of uninfected patients develop a fever. We then observe that the patient has developed a fever. How likely is the patient infected with the virus then?

Knowledge:

- Let Fev be the proposition “The person has a fever”
- Inf is a causal factor of both Fev and $Test$
- $P(Fev \mid Inf) = 0.7, P(Fev \mid \neg Inf) = 0.3$
- By Baye’s rule,

$$P(Inf \mid Fev \wedge Test) \propto P(Fev \mid Test \wedge Inf)P(Inf \mid Test).$$

We need extra knowledge for $P(Fev \mid Test \wedge Inf)$.

Definition

An atom X is **independent** of another atom Y **conditioned on** a set of atoms $Z_1, \dots, Z_m$ if for any $y \in \{0,1\}$

$$P(X \mid Z_1, \dots, Z_m) \times P(Y \mid Z_1, \dots, Z_m) = P(X, Y \mid Z_1, \dots, Z_m)$$

Example. [exam] Consider a domain that consists of students and exams. There are three atoms:

Smart, WorkHard, GoodAnswer

We may have the following samples

Smart	WorkHard	GoodAnswer	$P(\text{Smart}, \text{WorkHard}, \text{GoodAnswer})$
0	0	0	0.55
0	0	1	0.01
0	1	0	0.04
0	1	1	0.2
1	0	0	0.08
1	0	1	0.06
1	1	0	0.01
1	1	1	0.05

Then $P(\text{Smart}) = 0.2, P(\text{HardWork}) = 0.3$.

$$P(\text{Smart}) \times P(\text{WorkHard}) =$$

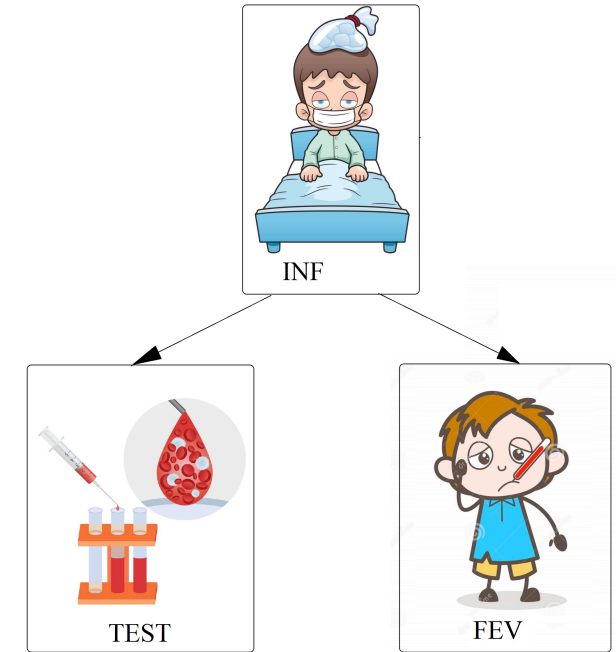
Smart	WorkHard	P
0	0	0.56
0	1	0.24
1	0	0.14
1	1	0.06

$$= P(\text{Smart}, \text{WorkHard}).$$

- So *Smart* and *WorkHard* are independent.
- But *Smart* and *WorkHard* are dependent conditioned on *GoodAnswer*.
- Suppose there is a fourth atom *GoodGrade*. *GoodGrade* only depends on the answer of the students. *GoodGrade* and *Smart* are not independent. But conditioned on *GoodAnswer*, *GoodGrade* and *Smart* are independent.

Example. [infection] Suppose we have an extra knowledge that *Fev* and *Test* are independent conditioned on *Inf*, i.e.,

- *Inf* is a causal factor of *Fev*, and *Test*
- *Fev* and *Test* are independent conditioned on *Inf*



We can represent this by clauses $Test \leftarrow Inf$ and $Fev \leftarrow Inf$.

<i>Inf</i>	$P(Inf)$	,	<i>Inf</i>	<i>Test</i>	$P(Test Inf)$	and	<i>Inf</i>	<i>Fev</i>	$P(Fev Inf)$
0	0.96		0	0	0.99		0	0	0.70
1	0.04		0	1	0.01		0	1	0.30
			1	0	0.05		1	0	0.30
			1	1	0.95		1	1	0.70

By conditional independence,

- $P(Fev | Test \wedge Inf) = P(Fev | Inf) = 0.70$
- $P(Fev | Test \wedge \neg Inf) = P(Fev | \neg Inf) = 0.30$

By Baye's theorem,

$$P(Inf | Fev \wedge Test) = \frac{P(Fev | Inf)P(Inf | Test)}{P(Fev | Test)} \approx 0.5588/P(Fev | Test)$$

$$P(\neg Inf | Fev \wedge Test) = \frac{P(Fev | \neg Inf)P(\neg Inf | Test)}{P(Fev | Test)} \approx 0.0605/P(Fev | Test)$$

Thus

$$P(Inf | Fev \wedge Test) : P(\neg Inf | Fev \wedge Test) \approx 0.5588/0.0605$$

Furthermore, $P(Inf \wedge Fev | Test) + P(\neg Inf | Fev \wedge Test) = 1$.

Therefore, $P(Inf \wedge Fev | Test) = \frac{0.7 \times 0.7983}{0.7 \times 0.7983 + 0.3 \times 0.2017} \approx 90\%$.

Example. [wumpus world] Suppose the agent senses a breeze at both squares (1,2) and (2,1).

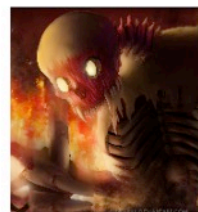
Propositions:

- $\text{Pit}_{i,j}, \text{Brz}_{i,j}$ for $(i,j) \in \{1,2,3,4\}^2$
- $\text{Known} = \neg \text{Pit}_{1,1} \wedge \neg \text{Pit}_{1,2} \wedge \neg \text{Pit}_{2,1}$
- $\text{Percept} = \neg \text{Brz}_{1,1} \wedge \text{Brz}_{1,2} \wedge \text{Brz}_{2,1}$

Assumptions:

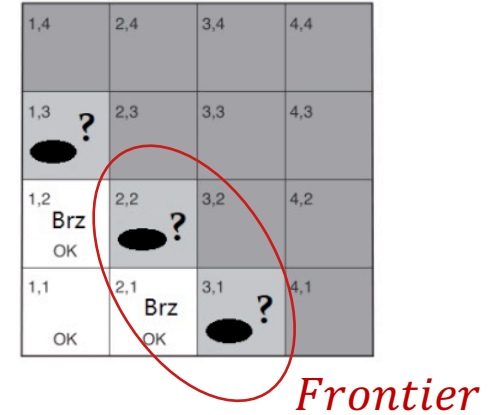
- A pit causes breezes in all adjacent squares, e.g., $P(\text{Brz}_{1,2} \mid \text{Pit}_{2,2}) = 1$
- Prior: $P(\text{Pit}_{i,j}) = 0.2$ for any $(i,j) \in \{1,2,3,4\}^2$.
- Independence: $P(\text{Pit}_{i,j}) \times P(\text{Pit}_{i',j'}) = P(\text{Pit}_{i,j} \wedge \text{Pit}_{i',j'})$ for $(i',j') \neq (i,j)$

Query: What is $P(\text{Pit}_{1,3} \mid \text{Known}, \text{Percept})$?



1,4	2,4	3,4	4,4
1,3 ?	2,3	3,3	4,3
1,2 Brz OK	2,2 ?	3,2	4,2
1,1 OK	2,1 Brz OK	3,1 ?	4,1

Define the set of propositions $Frontier = \{Pit_{2,2}, Pit_{3,1}\}$



$$P(Pit_{1,3} \mid Known, Percept) \propto P(Percept, Pit_{1,3}, Known)$$

$$= \sum_{Frontier} P(Percept, Pit_{1,3}, Known, Frontier)$$

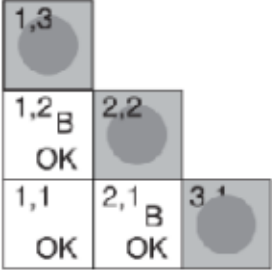
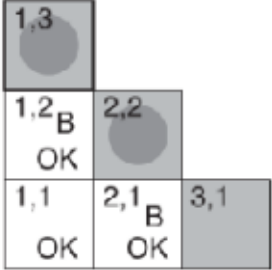
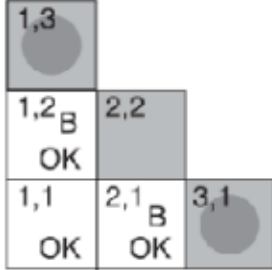
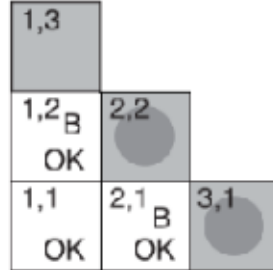
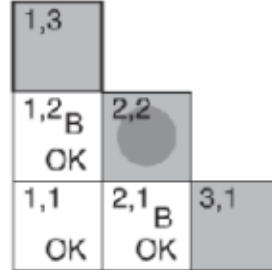
$$= \sum_{Frontier} P(Percept \mid Pit_{1,3}, Known, Frontier) P(Pit_{1,3}, Known, Frontier)$$

$$= \sum_{Frontier} P(Percept \mid Pit_{1,3}, Known, Frontier) P(Pit_{1,3}) P(Known) P(Frontier)$$

$$= P(Known) P(Pit_{1,3}) \sum_{Frontier} P(Percept \mid Pit_{1,3}, Known, Frontier) P(Frontier)$$

$$\propto P(Pit_{1,3}) \sum_{Frontier} P(Percept \mid Pit_{1,3}, Known, Frontier) P(Frontier)$$

To evaluate $P(\text{Percept} \mid \text{Pit}_{1,3}, \text{Known}, \text{Frontier})$ and $P(\text{Frontier})$:

					
$P(\text{Percept} \mid \text{Pit}_{1,3}, \text{Known}, \text{Frontier})$	1	1	1	1	1
$P(\text{Frontier})$	$0.2 \times 0.2 = 0.04$	$0.2 \times 0.8 = 0.16$	$0.8 \times 0.2 = 0.16$	$0.2 \times 0.2 = 0.04$	$0.2 \times 0.8 = 0.16$

Therefore

$$P(\text{Pit}_{1,3} \mid \text{Known}, \text{Percept}) \propto (0.2(0.04 + 0.16 + 0.16), 0.8(0.04 + 0.16)) \approx (0.31, 0.69)$$

Bayesian Networks



Diagnosing a Complex Problem

Example. [coughing?]

- When a person gets influenza (INF), he may develop a sore
- throat (SOR), a fever (FEV), or maybe bronchitis (BRO).
- When a person is a smoker (SMO), he may also get bronchitis.
- When a person gets bronchitis, he may develop coughing (COU)
- and wheezing (WEE).



Prior knowledge:

- INF and SMO are independent.
- SOR , FEV , and BRO are directly affected by INF , and are conditionally independent given INF . ($INF \rightarrow SOR, INF \rightarrow FEV, INF \rightarrow BRO$)
- BRO is also affected by SMO . ($SMO \rightarrow BRO$)
- COU and WEE are directly affected by BRO , and are conditionally independent given BRO . ($BRO \rightarrow COU, BRO \rightarrow WEE$)

For each atom, there is a conditional probability table:

INF	SMO	BRO	P_{BRO}
0	0	0	0.9999
0	0	1	0.0001
0	1	0	0.3
0	1	1	0.7
1	0	0	0.1
1	0	1	0.9
1	1	0	0.01
1	1	1	0.99

INF	P_{INF}
0	0.95
1	0.05

SMO	P_{SMO}
0	0.8
1	0.2

INF	SOR	P_{SOR}
0	0	0.9999
0	1	0.0001
1	0	0.7
1	1	0.3

INF	FEV	P_{FEV}
0	0	0.95
0	1	0.05
1	0	0.1
1	1	0.9

BRO	COU	P_{COU}
0	0	0.93
0	1	0.07
1	0	0.2
1	1	0.8

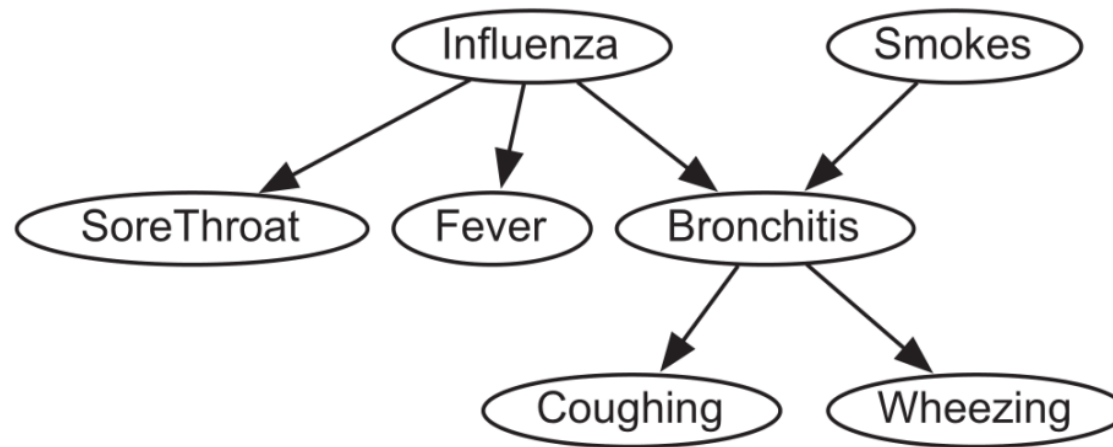
BRO	WEE	P_{WEE}
0	0	0.999
0	1	0.05
1	0	0.4
1	1	0.6

Question.

Suppose the person is coughing. What is the probability that he has influenza?

Probabilistic Model

The problem can be represented by a **directed graph** of the problem domain:



Note.

- The graph is acyclic (i.e. does not contain any directed cycles).
- A node X connects to another node Y by a directed edge if the X is a direct **causal factor** of Y .

Bayesian Networks

Definition [graph terminology]

Let $G = (V, E)$ be a directed graph.

- A **root** of G is any node with no parent;
- **$parents(X) = \{Y \in V \mid (Y, X) \in E\}$** is the set of parents of X .
- a **leaf** is any node that is not a parent of any others; the other nodes are called the **intermediate nodes**;
- A node X is a descendent of Y if there is a directed path from Y to X .

Definition [Bayesian network]

A **Bayesian network** is a tuple $(V, E, (P_X)_{X \in V})$ where

- (V, E) forms a directed acyclic graph; each node $X \in V$ represents an **atomic proposition**.
- For every node X , P_X is a **probability distribution of X conditioning on $parents(X)$** , i.e., $P(X \mid parents(X))$.
- **Local Markov property**: Each node is independent from its non-descendants conditioned on its parents.

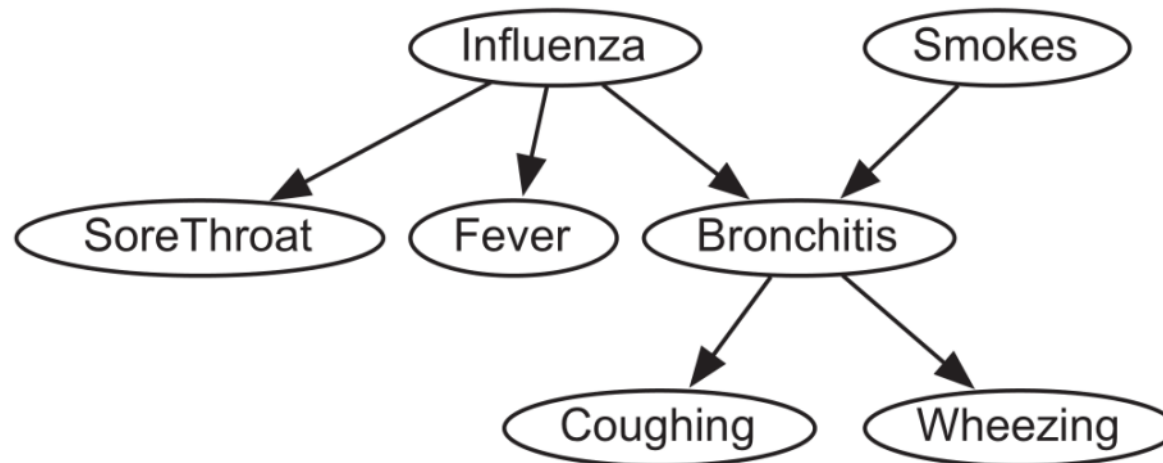
Example. [coughing?] A Bayesian network: $(V, E, (P_X)_{X \in V})$ where

- V contains atoms $INF, SMO, SOR, FEV, BRO, COU, WEE$
- (Syntax) E represents all the dependencies between atoms:

$INF \rightarrow SOR, INF \rightarrow FEV, INF \rightarrow BRO$
 $SMO \rightarrow BRO, BRO \rightarrow COU, BRO \rightarrow WEE$

- (Semantics) The CPT given earlier.

Note. Local Markov property ensures all observed conditional independence.



Bayesian Network Inference

For any atom X , we use l_X to denote a literal, i.e. X or $\neg X$.

Bayesian Network Inference Problem

INPUT Given a Bayesian network $(V, E, (P_X)_{X \in V})$, two nodes $X, Y \in V$, and literal $l_Y \in \{Y, \neg Y\}$,

OUTPUT compute the probability $P(X \mid l_X)$ and $P(\neg X \mid l_Y)$.

E.g. Given the earlier example Bayesian network, query for $P(INF \mid COU)$.

We now present methods to solve this problem.

Tools:

1. CPT operations
2. Chain rule

CPT Operations

CPT Operation 1: Restriction

For a conditional probability table $f(Y_1, \dots, Y_k, X)$, for $l_X \in \{X, \neg X\}$, the l_X -restriction operation produces a table $g(Y_1, \dots, Y_k)$ such that for any $l_{Y_i} \in \{Y_i, \neg Y_i\}$,

$$g(l_{Y_1}, \dots, l_{Y_k}) = f(l_{Y_1}, \dots, l_{Y_k}, l_X).$$

E.g.

Y	X	$f(Y, X)$	$X = 1$ $\Rightarrow$	Y	$g(Y)$
0	0	0.99			
0	1	0.01		0	0.01
1	0	0.05		1	0.95
1	1	0.95			

CPT Operations

CPT Operation 2: Multiplication

For CPT $f(Y_1, \dots, Y_k, X)$ and $g(X, Z_1, \dots, Z_m)$, the **multiplication** operation produces a new CPT $h(Y_1, \dots, Y_k, X, Z_1, \dots, Z_m)$ such that for any literals $l_{Y_i} \in \{Y_i, \neg Y_i\}$, $l_X \in \{X, \neg X\}$, $l_{Z_j} \in \{Z_j, \neg Z_j\}$,

$$h(l_{Y_1}, \dots, l_{Y_k}, l_X, l_{Z_1}, \dots, l_{Z_m}) = f(l_{Y_1}, \dots, l_{Y_k}, l_X) \times g(l_X, l_{Z_1}, \dots, l_{Z_m}).$$

E.g.

A	B	f		B	C	g		A	B	C	h
0	0	0.1		0	0	0.2		0	0	0	0.02
0	1	0.9		0	1	0.8		0	0	1	0.08
1	0	0.4	$\times$	1	0	0.3	$=$	0	1	0	0.27
1	1	0.6		1	1	0.7		0	1	1	0.63
								1	0	0	0.08
								1	0	1	0.32
								1	1	0	0.18
								1	1	1	0.42

CPT Operations

CPT Operation 3: Normalisation

For a conditional probability table $f(Y_1, \dots, Y_k, X)$, the **X -normalisation** operation produces a new CPT **$g(Y_1, \dots, Y_k, X)$** such that for any $l_{Y_i} \in \{Y_i, \neg Y_i\}$, $l_X \in \{X, \neg X\}$,

$$g(l_{Y_1}, \dots, l_{Y_k}, l_X) = \frac{f(l_{Y_1}, \dots, l_{Y_k}, l_X)}{f(l_{Y_1}, \dots, l_{Y_k}, X) + f(l_{Y_1}, \dots, l_{Y_k}, \neg X)}.$$

E.g.

X	$f(X)$	$\Rightarrow$	X	$g(X)$
0	0.0096		0	0.2017
1	0.038		1	0.7983

CPT Operations

CPT Operation 4: Summation

For a conditional probability table $f(Y_1, \dots, Y_k, X)$, the **X-sum** operation produces a new CPT $g(Y_1, \dots, Y_k)$ such that for any $l_{Y_i} \in \{Y_i, \neg Y_i\}$,

$$g(l_{Y_1}, \dots, l_{Y_k}) = f(l_{Y_1}, \dots, l_{Y_k}, X) + f(l_{Y_1}, \dots, l_{Y_k}, \neg X).$$

E.g. C-sum of the following table:

A	B	C	f
0	0	0	0.02
0	0	1	0.72
0	1	0	0.27
0	1	1	0.63
1	0	0	0.08
1	0	1	0.32
1	1	0	0.18
1	1	1	0.42



Y	X	g
0	0	0.74
0	1	0.9
1	0	0.4
1	1	0.6

Example. [simple diagnosis] To query $P(Inf|Test)$

Inf	P_{Inf}
0	0.96
1	0.04

and

Inf	Test	$P_{Test}(Inf)$
0	0	0.99
0	1	0.01
1	0	0.05
1	1	0.95

Note. $P(Inf|Test) \propto P(Inf) \times P(Test|Inf)$

Step 1 Restriction.

In P_{Test} keep only those rows where $Test$ is true.

Inf	$f(Inf)$
0	0.01
1	0.95

Step 2 Multiplication.

Inf	P_{Inf}		Inf	$f(Inf)$		Inf	$f(Inf)$
0	0.96	$\times$	0	0.01	$=$	0	$0.01 \times 0.96 = 0.0096$
1	0.04		1	0.95		1	$0.95 \times 0.04 = 0.038$

Step 3 Normalisation.

Inf	$g(Inf)$
0	$0.0096 / (0.0096 + 0.038) = 0.0096 / 0.0476 \approx 0.2017$
1	$0.038 / (0.0096 + 0.038) = 0.038 / 0.0476 \approx 0.7983$

Therefore $P(Inf|Test) = 0.7983$ and $P(\neg Inf|Test) = 0.2017$.

Chain Rule

- For any two propositions a_1, a_2

$$P(a_1 \wedge a_2) = P(a_2|a_1)P(a_1).$$

- For any three propositions a_1, a_2, a_3

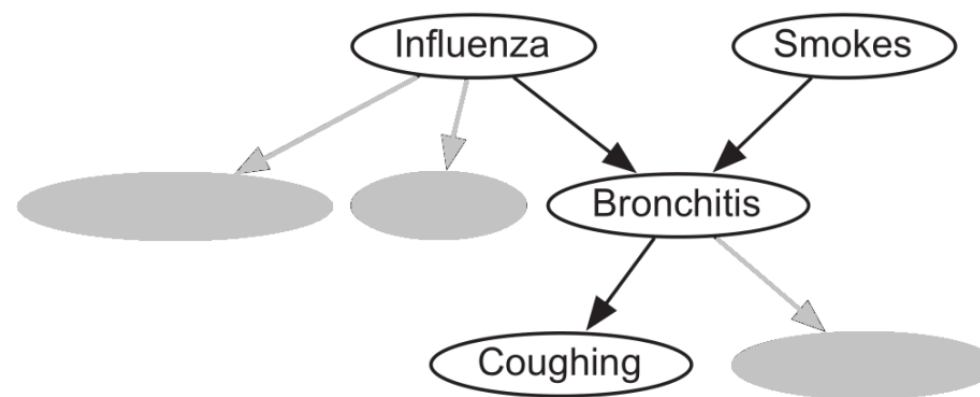
$$P(a_1 \wedge a_2 \wedge a_3) = P(a_1 \wedge a_2|a_3)P(a_3) = P(a_1|a_2 \wedge a_3)P(a_2|a_3)P(a_3).$$

Chain Rule

For any n proposition $a_1, a_2, \dots, a_n$

$$\begin{aligned} P\left(\bigwedge_{i=1}^n a_i\right) &= P(a_1|a_2 \wedge \dots \wedge a_n)P(a_2|a_3 \wedge \dots \wedge a_n) \dots P(a_{n-1}|a_n)P(a_n) \\ &= \prod_{i=1}^n P\left(a_i \mid \bigwedge_{j=i+1}^n a_j\right) \end{aligned}$$

Example. [coughing?]



$$\begin{aligned} &P(INF, SMO, BBRO, COU) \\ &= P(COU|BRO, SMO, INF)P(BRO|SMO, INF)P(SMO|INF)P(INF) \\ &= P(COU|BRO)P(BRO|SMO, INF)P(SMO)P(INF) \quad (\text{by local Markov property}) \end{aligned}$$

Theorem [chain rule on a Bayesian network]

Let $(V, E, (P_X)_{X \in V})$ be a Bayesian network. Then for any subset $s \subseteq V$

$$P(S) = \prod_{X \in S} P(X|parents(X)) = \prod_{X \in S} P_X$$

Example. [coughing?] what is $P(INF|COU)$?

Idea:

$$P(INF|COU) = P(COU|INF)P(INF) = P(COU, INF)$$

$$= \sum_{BRO} \sum_{SMO} P(INF, COU, BRO, SMO)$$

$$= \sum_{BRO} \sum_{SMO} P_{COU} P_{BRO} P_{SMO} P_{INF}$$

Starting from the set of CPT: $P_{COU}, P_{BRO}, P_{SMO}, P_{INF}$

INF	SMO	BRO	P_{BRO}
0	0	0	0.9999
0	0	1	0.0001
0	1	0	0.3
0	1	1	0.7
1	0	0	0.1
1	0	1	0.9
1	1	0	0.01
1	1	1	0.99

INF	P_{INF}
0	0.95
1	0.05

SMO	P_{SMO}
0	0.8
1	0.2

BRO	COU	P_{COU}
0	0	0.93
0	1	0.07
1	0	0.2
1	1	0.8

Question. How to compute $\sum_{BRO} \sum_{SMO} P_{COU} P_{BRO} P_{SMO} P_{INF}$

Naïve method.

- **Step 1: Restriction.** In P_{COU} , keep only those rows where COU is true.
Obtain a new CPT $f(BRO)$.
- **Step 2: Multiplication.** Compute $f(BRO) \times P_{BRO} \times P_{SMO} \times P_{INF}$
Obtain a new CPT $g(BRO, SMO, INF)$
- **Step 3: Sum (to eliminate variables).** Perform SMO -sum and then BRO -sum, on $g(BRO, SMO, INF)$.
Obtain a new CPT $h(INF)$
- **Step 4: Normalisation.** Extract the conditional probabilities $P(INF|COU)$ and $P(\neg INF|COU)$.

Complexity. Suppose a CPT has r rows. The algorithm runs in $O(r^n)$.

We now present how to improve the running time of the algorithm.

Variable Elimination

Idea: Instead of multiply all tables and eliminate variables all at once, eliminate variables one-by-one.

- **Step 1: Restriction.** In P_{COU} , keep only those rows where COU is true.
Obtain a new CPT $f(BRO)$.
- **Step 2: Eliminate SMO .**
 - **Step 2.1:** Multiply $P_{BRO} \times P_{SMO}$ to get a new table $g(INF, SMO, BRO)$
 - **Step 2.2:** SMO -sum $g(INF, SMO, BRO)$ to get a new table $h(INF, BRO)$.
- **Step 3: Eliminate BRO .**
 - **Step 3.1:** Multiply $f(BRO) \times h(INF, BRO)$ to get a new table $l(INF, BRO)$
 - **Step 3.2:** BRO -sum $l(INF, BRO)$ to get a new table $r(INF)$.
- **Step 4: Multiplication.** $r(INF) \times P(INF)$ to a table $s(INF)$.
- **Step 5: Normalisation.** Retrieve $P(INF|COU)$.

Illustration

Step 1: Restriction.

In P_{COU} , keep only those rows where COU is true.

BRO	COU	P_{COU}
0	0	0.93
0	1	0.07
1	0	0.2
1	1	0.8

 $\Rightarrow$

BRO	$f(BRO)$
0	0.07
1	0.8

Step 2:
 Multiplication *SMO*.

2.1

INF	SMO	BRO	P_{BRO}				INF	SMO	BRO	$g(INF, SMO, BRO)$						
0	0	0	0.9999				0	0	0	0.79992						
0	0	1	0.0001				0	0	1	0.00008						
0	1	0	0.3				0	1	0	0.06						
0	1	1	0.7	×	<table><tr><th>SMO</th><th>P_{SMO}</th></tr><tr><td>0</td><td>0.8</td></tr><tr><td>1</td><td>0.2</td></tr></table>	SMO	P_{SMO}	0	0.8	1	0.2	=	0	1	1	0.14
SMO	P_{SMO}															
0	0.8															
1	0.2															
1	0	0	0.1				1	0	0	0.08						
1	0	1	0.9				1	0	1	0.72						
1	1	0	0.01				1	1	0	0.002						
1	1	1	0.99				1	1	1	0.198						

2.2 *SMO*-Sum

INF	SMO	BRO	$g(INF, SMO, BRO)$			INF	BRO	$h(INF, BRO)$
0	0	0	0.79992			0	0	0.85992
0	0	1	0.00008			0	1	0.14008
0	1	0	0.06			1	0	0.082
0	1	1	0.14			1	1	0.918
1	0	0	0.08					
1	0	1	0.72					
1	1	0	0.002					
1	1	1	0.198					

⇒

INF	BRO	$h(INF, BRO)$
0	0	0.85992
0	1	0.14008
1	0	0.082
1	1	0.918

Step 3: Eliminate BRO .

3.1

BRO	P_{COU}
0	0.07
1	0.8

×

INF	BRO	$h(INF, BRO)$
0	0	0.85992
0	1	0.14008
1	0	0.082
1	1	0.918

=

INF	BRO	$l(INF, BRO)$
0	0	0.0601944
0	1	0.112064
1	0	0.00574
1	1	0.7344

3.2 BRO -Sum

INF	BRO	$l(INF, BRO)$
0	0	0.0601944
0	1	0.112064
1	0	0.00574
1	1	0.7344

⇒

INF	$r(INF)$
0	0.1722584
1	0.74014

Step 4: Multiplication.

INF	$r(INF)$
0	0.1722584
1	0.74014

×

INF	P_{INF}
0	0.95
1	0.05

=

	$s(INF)$
0	0.16364548
1	0.037007

Step 5: Normalisation.

	$s(INF)$
0	0.16364548
1	0.037007

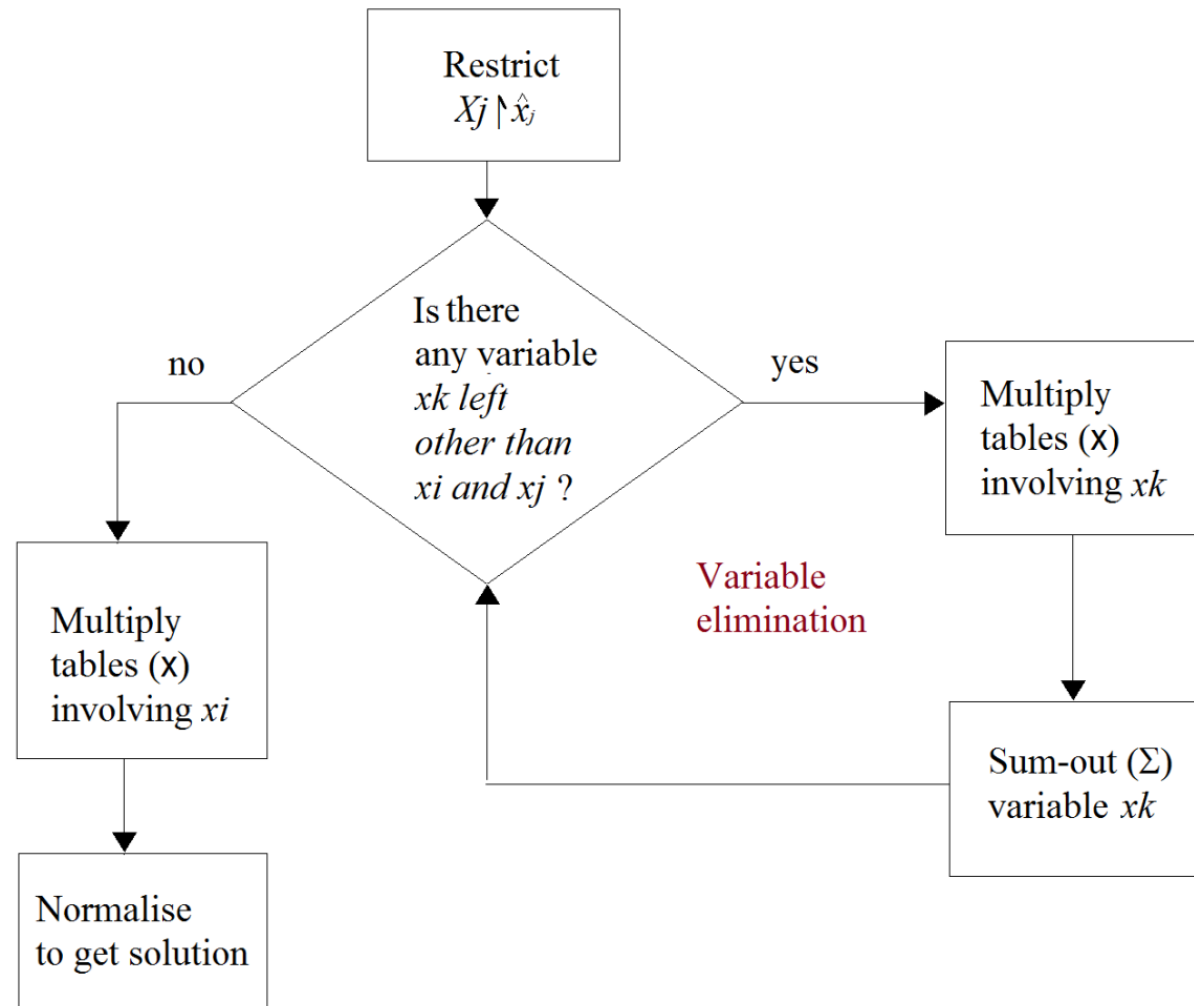
⇒

	Result
$P(\neg INF \wedge COU)$	0.8156
$P(INF \wedge COU)$	0.1844

The probability of a person getting influenza when (only) coughing is observed is 18.44%!

Variable Elimination Algorithm

Solving the Bayesian network inference problem using the **variable elimination algorithm (VE)**



Variable Elimination (Rough steps)

INPUT:

- CPT P_X for each $X \in V$.
- Query variables X_i, X_j ; Observed value for X_j .

OUTPUT: $P(X_i|X_j)$

1. Restrict tables that contain X_j to rows where X_j has the observed value.
2. Eliminate each of the non-query variables X_k (in certain order):
 - Multiply all tables containing X_k .
 - Sum-out X_k from the product table.
3. Multiply the remaining tables to get a table over a single variable X_i .
4. Normalise and return the result.

Complexity: If every atom appears in at most 2 CPT and one of them has at most 4 rows, then the algorithm runs in $O(n)$.

In the general case, the running time depends on the ordering in which we eliminate the variables, but finding the optimal ordering is NP-complete.

Summary of The Topic

The following are the main knowledge points covered:

- Probability theory
- Conditional probability and its properties
 - Chain rule
 - Law of total probability
 - Baye's rule
- Independence and conditional independence
- Bayesian Network
 - Local Markov property
 - Variable elimination